

|                                                                                                                                                                                                                           |  |  |  |                                       |  |  |  |                                                                  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|---------------------------------------|--|--|--|------------------------------------------------------------------|--|
| CLIENT<br>CUSTOMER                                                                                                                                                                                                        |  |  |  | LIEU D'INSTALLATION<br>PLANT LOCATION |  |  |  | REV                                                              |  |
| FONCTION<br>SERVICE                                                                                                                                                                                                       |  |  |  | RECYCLED WATER PUMP (10 KG/CM2G)      |  |  |  | QUANTITE NECESSAIRE<br>N° OF UNITS                               |  |
| ENTRAINE PAR<br>DRIVEN BY                                                                                                                                                                                                 |  |  |  | ELECTRIC MOTOR                        |  |  |  | CONTINU<br>CONTINUOUS                                            |  |
| NOM DU PROJET<br>PROJECT NAME                                                                                                                                                                                             |  |  |  | TYPE                                  |  |  |  | DISCONTINU<br>DISCONTINUOUS                                      |  |
| REFERENCE OFFRE CONSTRUCTEUR<br>MFR PROPOSAL N°                                                                                                                                                                           |  |  |  | TYPE                                  |  |  |  | Centrifugal Pump                                                 |  |
| CONDITIONS OPERATOIRES<br>OPERATING CONDITIONS                                                                                                                                                                            |  |  |  |                                       |  |  |  | CARACTERISTIQUES<br>PERFORMANCES                                 |  |
| 1 LIQUID<br>LIQUID                                                                                                                                                                                                        |  |  |  |                                       |  |  |  | 2 RECYCLED WATER                                                 |  |
| 3 DEBIT Nominal<br>FLOW Nominal                                                                                                                                                                                           |  |  |  |                                       |  |  |  | 100 DEBIT Maxi.<br>FLOW Maxi.                                    |  |
| 4 PRESSION/TEMP. DE REF.<br>PRESSURE/TEMP. REF                                                                                                                                                                            |  |  |  |                                       |  |  |  | 65 °C (5)                                                        |  |
| 5 MASSE VOL. A PT<br>DENSITY AT PT                                                                                                                                                                                        |  |  |  |                                       |  |  |  | 981 kg/m3                                                        |  |
| 6 TENSION DE VAPEUR A T<br>VAPOR PRESSURE AT T                                                                                                                                                                            |  |  |  |                                       |  |  |  | 0.25 kg/cm²a                                                     |  |
| 7 VISCOSITE A PT<br>VISCOSITY AT PT                                                                                                                                                                                       |  |  |  |                                       |  |  |  | 0.4 cP                                                           |  |
| 8 CORROSION/EROSION DUE A<br>CORROSION/EROSION CAUSED BY                                                                                                                                                                  |  |  |  |                                       |  |  |  | -                                                                |  |
| 9 TYPE ET MATERIAUX DE CONSTRUCTION<br>CONSTRUCTION AND MATERIALS                                                                                                                                                         |  |  |  |                                       |  |  |  | SENS DE ROTATION FACE ACCOUP.<br>ROTATION (VIEWED FROM CPLG END) |  |
| 10 SUPPORTAGE CORPS<br>CASING MOUNTING                                                                                                                                                                                    |  |  |  |                                       |  |  |  | AXIAL<br>CENTERLINE                                              |  |
| 11 JOINT<br>SPLIT                                                                                                                                                                                                         |  |  |  |                                       |  |  |  | HORIZONTAL<br>AXIAL                                              |  |
| 12 TYPE<br>TYPE                                                                                                                                                                                                           |  |  |  |                                       |  |  |  | VOLUTE SIMPLE<br>SINGLE VOLUTE                                   |  |
| 13 BOSSAGES<br>TAPPED OPENINGS                                                                                                                                                                                            |  |  |  |                                       |  |  |  | EVENT<br>VENT                                                    |  |
| 14 TUBULURES<br>NOZZLES                                                                                                                                                                                                   |  |  |  |                                       |  |  |  | D.N.<br>SIZE                                                     |  |
| 15 ASPIRATION<br>SUCTION                                                                                                                                                                                                  |  |  |  |                                       |  |  |  | (3)                                                              |  |
| 16 REFOULEMENT<br>DISCHARGE                                                                                                                                                                                               |  |  |  |                                       |  |  |  | (3)                                                              |  |
| 17 DIAMETER DE LA ROUE<br>IMPELLER DIAMETER                                                                                                                                                                               |  |  |  |                                       |  |  |  | MAX<br>MAX                                                       |  |
| 18 PALIER<br>BEARING                                                                                                                                                                                                      |  |  |  |                                       |  |  |  | SOCLE<br>BASE                                                    |  |
| 19 ACCOUPLET MARQUE<br>COUPLING MFR                                                                                                                                                                                       |  |  |  |                                       |  |  |  | TYPE<br>TYPE                                                     |  |
| 20 PROTECTEUR D'ACCOUPLEMENT<br>COUPLING GUARD                                                                                                                                                                            |  |  |  |                                       |  |  |  | Anti-sparking                                                    |  |
| 21 PRESSE-ETOUPE (P.E.)<br>PACKING TYPE AND SIZE                                                                                                                                                                          |  |  |  |                                       |  |  |  | AVEC LANTERNE<br>WITH LANTERN                                    |  |
| 22 GARNITURE MECANIQUE (G.M.)<br>MECHANICAL SEAL MFR                                                                                                                                                                      |  |  |  |                                       |  |  |  | (12)                                                             |  |
| 23 G.M. DIAM. ET CARACTERISTIQUES<br>M.S. SIZE AND CHARACTERISTICS                                                                                                                                                        |  |  |  |                                       |  |  |  | TYPE<br>TYPE                                                     |  |
| 24 G.M. DESIGNATION SELON API 610<br>M.S. CLASSIFICATION CODE ACCORDING TO API 610                                                                                                                                        |  |  |  |                                       |  |  |  | Single<br>Seal Plan 11                                           |  |
| 25 REMARQUES<br>REMARKS                                                                                                                                                                                                   |  |  |  |                                       |  |  |  | General Note : Pump to be insulated for personnel protection.    |  |
| 26 (1) VALUE STATED FOR DESIGN FLOWRATE.                                                                                                                                                                                  |  |  |  |                                       |  |  |  |                                                                  |  |
| 27 (2) DELETED.                                                                                                                                                                                                           |  |  |  |                                       |  |  |  |                                                                  |  |
| 28 (3) TO BE PROVIDED BY VENDOR.                                                                                                                                                                                          |  |  |  |                                       |  |  |  |                                                                  |  |
| 29 (4) ESTIMATED SHAFT POWER TO BE CONFIRMED BY PUMP VENDOR. MOTOR SHALL BE ENERGY-EFFICIENT : CEI60034-30, CLASS IE2 OR IE3, OR EQUIVALENT NORM. TO BE CONFIRMED BY ELECTRICAL DISCIPLINE IN MECHANICAL DATASHEET (MDS). |  |  |  |                                       |  |  |  | 01                                                               |  |
| 30 REFER TO PAGE 3/3 FOR THE OTHER NOTES                                                                                                                                                                                  |  |  |  |                                       |  |  |  |                                                                  |  |
| POMPE CENTRIFUGE<br>CENTRIFUGAL PUMP                                                                                                                                                                                      |  |  |  | REPERE - ITEM                         |  |  |  | PAGE                                                             |  |
| MP-PP031-P600173A/B                                                                                                                                                                                                       |  |  |  | MP-PP031-P700173A/B                   |  |  |  | DATE                                                             |  |
|                                                                                                                                                                                                                           |  |  |  |                                       |  |  |  | REV                                                              |  |
|                                                                                                                                                                                                                           |  |  |  |                                       |  |  |  | 00                                                               |  |
|                                                                                                                                                                                                                           |  |  |  |                                       |  |  |  | 01                                                               |  |

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| CLIENT<br>CUSTOMER                                                                                                                                                                       |  |                                        |  | JEU D'INSTALLATION<br>PLANT LOCATION                  |  |                                                   |  | REV                                 |  |
| FONCTION<br>SERVICE                                                                                                                                                                      |  |                                        |  | RECYCLED WATER PUMP (10 KG/CM2G)                      |  |                                                   |  | QUANTITE NECESSAIRE<br>N° OF UNITS  |  |
| ENTRAINE PAR<br>DRIVEN BY                                                                                                                                                                |  |                                        |  | ELECTRIC MOTOR                                        |  |                                                   |  | CONTINU<br>CONTINUOUS               |  |
| NOM DU PROJET<br>PROJECT NAME                                                                                                                                                            |  |                                        |  | SERVICE<br>OPERATION                                  |  |                                                   |  | DISCONTINU<br>DISCONTINUOUS         |  |
| REFERENCE OFFRE CONSTRUCTEUR<br>MFR PROPOSAL N°                                                                                                                                          |  |                                        |  | TYPE<br>TYPE                                          |  |                                                   |  | Centrifugal Pump                    |  |
| 31 MATERIAUX POMPE<br>MATERIALS PUMP                                                                                                                                                     |  | GARNITURE MECANIQUE<br>MECHANICAL SEAL |  | ESSAIS CHEZ CONSTRUCTEUR<br>SHOP TESTS                |  | DEMANDE<br>REQUIRED                               |  | CONTR. CLIENT<br>PURCH. CHECK       |  |
| 32 ROUE<br>IMPELLER                                                                                                                                                                      |  | SS 304 (6)                             |  | GRAIN FIXE COTE P.<br>STAT. SEAL PUMP SIDE            |  | PERFORMANCE<br>PERFORMANCE                        |  | YES YES                             |  |
| 33 CORPS<br>INNER CASE                                                                                                                                                                   |  | SS 304 (6)                             |  | GRAIN TOURN. COTE P.<br>ROT. SEAL. PUMP SIDE          |  | NPSH<br>NPSH                                      |  |                                     |  |
| 34 CHEMISE (P.E.)<br>SLEEVE (PACKED)                                                                                                                                                     |  |                                        |  | SOUFFLET<br>BELLOWS                                   |  | SS                                                |  | DEMONTAGE<br>DISMANTLING            |  |
| 35 CHEMISE (G.M.)<br>SLEEVE (M.S.)                                                                                                                                                       |  | SS 304 (6)                             |  | AUTRES PIECES<br>OTHERS                               |  | SS                                                |  | PROCES-VERBAL<br>TEST RECORD        |  |
| 36 BAGUE D'USURE<br>WEAR RINGS                                                                                                                                                           |  |                                        |  | GRAIN FIXE COTE EXT.<br>STAT. SEAL EXT. SIDE          |  | EPREUVE HYDRAU.<br>HYDRAUL. TESTS                 |  | YES YES                             |  |
| 37 ARBRE<br>SHAFT                                                                                                                                                                        |  |                                        |  | GRAIN TOURN. C. EXT.<br>ROT. SEAL EXT. SIDE           |  | P/T MAX. ADMISSIBLE<br>P/T MAX. ALLOWABLE         |  | kg/cm²g °C                          |  |
| 38 JOINTS DE CORPS<br>CASING GASKET                                                                                                                                                      |  |                                        |  | JOINTS<br>GASKETS & O RINGS                           |  | POIDS POMPE<br>WEIGHT PUMP                        |  | kg SOCLE<br>BASE kg                 |  |
| 39                                                                                                                                                                                       |  |                                        |  | POT<br>RESERVOIR                                      |  | LIEU DES ESSAIS<br>TESTS LOCATION                 |  | At vendor workshop                  |  |
| 40                                                                                                                                                                                       |  |                                        |  | LIQUIDE TAMPON<br>BUFFING LIQUID                      |  |                                                   |  |                                     |  |
| 41 MOTEUR FOURNI PAR<br>MOTOR SUPPLIED BY                                                                                                                                                |  | ENTRAINEMENT<br>DRIVERS                |  | CARACTERISTIQUES REELLES<br>MFR FINAL DATA (AS BUILT) |  |                                                   |  |                                     |  |
| 42 MARQUE<br>MFR                                                                                                                                                                         |  | TYPE<br>TYPE                           |  | ACCOUPLEMENT<br>COUPLING                              |  | DIAM. REEL ROUE<br>ACTUAL IMPELLER DIAMETER       |  |                                     |  |
| 43 kW<br>kW                                                                                                                                                                              |  | Tr/min<br>RPM                          |  | Carc.<br>Frame                                        |  | Fte                                               |  | POULIES/COURROIES<br>V BELT/PULLEYS |  |
| 44 PROTECTION<br>PROTECTION                                                                                                                                                              |  | IP65                                   |  |                                                       |  | PLAN D'ENSEMBLE N°<br>OUTLINE DRWG N°             |  |                                     |  |
| 45 ANTIDÉFLAGRANT<br>EXPLOSION PROOF                                                                                                                                                     |  | (13)                                   |  |                                                       |  | PLAN EN COUPE N°<br>SECTIONAL DRWG N°             |  |                                     |  |
| 46 VOLTS/PHASES/Hz<br>VOLTS/PHASES/Hz                                                                                                                                                    |  | 415/3/50                               |  | VARIATION DE VITESSE<br>VARIABLE SPEED DRIVE UNIT     |  | PLAN DE LA GARNITURE N°<br>SEALING SYSTEM DRWG N° |  |                                     |  |
| 47 PALIER<br>BRGS                                                                                                                                                                        |  | LUB.<br>LUB.                           |  | Grease                                                |  | N° DE LA SERIE DE LA POMPE<br>PUMP SERIAL N°      |  |                                     |  |
| 48 INTENSITE A CHARGE MAXI<br>CURRENT AT MAX LOADING                                                                                                                                     |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 49 MONTE PAR<br>MOUNTED BY                                                                                                                                                               |  | POIDS<br>WT                            |  | kg                                                    |  |                                                   |  |                                     |  |
| 50 REMARQUES<br>REMARKS                                                                                                                                                                  |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 51 (5) DESIGN PRESSURE: SUCTION DESIGN PRESSURE = 0.68 kg/cm2g. DESIGN TEMPERATURE = 98°C.                                                                                               |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 52 (6) STAINLESS STEEL WITH CARBON CONTENT LESS THAN 0.04%                                                                                                                               |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 53 (7) MDMT = 5°C                                                                                                                                                                        |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 54 (8) MOTOR RE-ACCELERATION REQUIREMENT TO BE CONSIDERED.                                                                                                                               |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 55 (9) NPSHA INDICATED IS FROM PUMP CENTERLINE WHICH IS AT 800 mm ELEVATION FROM GRADE.                                                                                                  |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 56 (10) MOTOR SHALL BE SELECTED FOR END OF THE CURVE.                                                                                                                                    |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 57 (11) ESTIMATED DISCHARGE DESIGN PRESSURE (CORRESPONDING TO SHUT-OFF PRESSURE) IS 16.5 kg/cm2g. VENDOR SHALL LIMIT RISE TO SHUT-OFF SO AS NOT TO EXCEED THE DISCHARGE DESIGN PRESSURE. |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 58 (12) SEAL PLAN AND FLUSHING LIQUID REQUIREMENT FOR PUMP TO BE CONFIRMED BY MECHANICAL DISCIPLINE IN MDS BASED ON VENDOR REQUIREMENT.                                                  |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 59 (13) MOTOR TYPE BASED ON HAZARDOUS AREA CLASSIFICATION TO BE CONFIRMED BY ELECTRICAL AND HSE DISCIPLINE IN MDS.                                                                       |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
| 60                                                                                                                                                                                       |  |                                        |  |                                                       |  |                                                   |  |                                     |  |
|                                                                                                                                                                                          |  | POMPE CENTRIFUGE<br>CENTRIFUGAL PUMP   |  |                                                       |  | PAGE                                              |  | DATE                                |  |
|                                                                                                                                                                                          |  | REPERE - ITEM                          |  | MP-PP031-P600173A/B<br>MP-PP031-P700173A/B            |  | 3 / 3                                             |  | 26/06/2024                          |  |
|                                                                                                                                                                                          |  |                                        |  |                                                       |  |                                                   |  | REV                                 |  |
|                                                                                                                                                                                          |  |                                        |  |                                                       |  |                                                   |  | 00                                  |  |
|                                                                                                                                                                                          |  |                                        |  |                                                       |  |                                                   |  | 01                                  |  |